Supporting Information

Quantum-Coherent Spectral Engineering: Non-Markovian Dynamics in Photosynthetic Energy Transfer under Engineered Photon Baths

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May 3, 2026

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Associated Manuscript:
This Supporting Information accompanies the manuscript submitted to the Journal of Physical Chemistry Letters. All simulations are performed using the Process Tensor Hierarchy of Pure States (PT-HOPS) method with Stochastically Bundled Dissipators (SBD) as implemented in the MesoHOPS framework[?,?].

Contents

Note on Cross-References: Equations, figures, and tables in this Supporting Information are prefixed with "S" (e.g., Eq. S1, Fig. S1, Table S1). References to the main manuscript are indicated as "main text" or by unadorned numbers (e.g., Eq. 1, Fig. 1).

S1 PT-HOPS/SBD formalism details

S1.1 System Hamiltonian and spectral filtering

The reduced density matrix $\boldsymbol{\rho}(t)$ evolves according to:

$$\frac{\partial \boldsymbol{\rho}(t)}{\partial t} = -\frac{i}{\hbar}\left[\hat{H}_S, \boldsymbol{\rho}(t)\right] + \mathcal{D}[\boldsymbol{\rho}(t)], \tag{S1}$$

where $\mathcal{D}$ represents the dissipator capturing system-bath interactions beyond the Markovian approximation[?] . The electronic Hamiltonian reads:

$$\hat{H}_{\mathrm{el}} = \sum_{n=1}^{7} \varepsilon_n \left|n\right\rangle\!\left\langle n\right| + \sum_{n\neq m} J_{nm} \left|n\right\rangle\!\left\langle m\right|, \tag{S2}$$

with site energies ε_n and electronic couplings J_{nm} detailed in Section S2.

We consider transient dynamics following impulsive excitation by a spectrally shaped femtosecond laser pulse. The effective driving field spectrum determining the state preparation is:

$$E_{\mathrm{eff}}(\omega) = T(\omega)\cdot E_{\mathrm{in}}(\omega), \tag{S3}$$

where $E_{\mathrm{in}}(\omega)$ is the incident broadband pulse spectrum, and $T(\omega)$ is the transmission function of the optical pulse shaper:

$$T(\omega) = T_{\mathrm{peak}} \sum_{i=1}^{n} w_i \exp\left[-\frac{(\omega-\omega_{c,i})^2}{2\sigma_i^2}\right], \tag{S4}$$

with $n = 2$ for dual-band filtering targeting the selected vibronic resonances as analyzed in 2D electronic spectroscopy[?] .

The incident broadband pulse has a Gaussian temporal envelope:

$$E(t) = E_0 \exp\left(-\frac{t^2}{2\sigma_t^2}\right), \tag{S5}$$

where E_0 is the peak field amplitude, $\sigma_t = \mathrm{FWHM}/(2\sqrt{2\ln 2})$, and the pulse is centered at $t = 0$ relative to the start of the dynamical integration. All simulations use FWHM $= 50\,\mathrm{fs}$ (corresponding to $\sigma_t \approx 21.2\,\mathrm{fs}$ and a spectral bandwidth $\Delta\omega \approx 294\,\mathrm{cm}^{-1}$), which is broader than the filter transmission peaks (FWHM $\approx 300\,\mathrm{cm}^{-1}$), ensuring the filter—not the pulse—determines the spectral content of the initial state.

S1.2 Hierarchy equations

The Process Tensor Hierarchy of Pure States (PT-HOPS) method extends the standard HOPS formalism by incorporating a process tensor that efficiently captures non-Markovian environmental memory. The hierarchy of pure states $\left|\psi^{(\mathbf{n})}(t)\right\rangle$ evolves according to:

$$\frac{\partial}{\partial t}\left|\psi^{(\mathbf{n})}(t)\right\rangle = \left(-i\mathbf{H}_S - \sum_j n_j \nu_j\right)\left|\psi^{(\mathbf{n})}(t)\right\rangle - i\sum_j \sqrt{(n_j+1)d_j}\,\hat{L}_j\left|\psi^{(\mathbf{n}+\mathbf{e}_j)}(t)\right\rangle - i\sum_j \sqrt{n_j/d_j}\,\hat{L}_j^{\dagger}\left|\psi^{(\mathbf{n}-\mathbf{e}_j)}(t)\right\rangle \tag{S6}$$

where we follow the standard Ishizaki–Fleming derivation for the hierarchy[?][?] , and where:

- $\mathbf{n} = (n_1, n_2, \ldots, n_K)$ is the multi-index for hierarchy level
- ν_j are the bath correlation frequencies from Padé decomposition
- d_j are the decomposition coefficients
- $\hat{L}_j$ are the system coupling operators
- $\mathbf{e}_j$ is the unit vector in direction j

S1.3 Bath correlation function decomposition

The bath correlation function is decomposed using Padé spectral decomposition:

$$C(t) = \sum_{k=0}^{K} d_k e^{-\nu_k t}, \tag{S7}$$

where K is the number of Matsubara terms. For the Drude–Lorentz spectral density:

$$J_{\mathrm{DL}}(\omega) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2}, \tag{S8}$$

the correlation function at temperature T is:

$$C(t) = \frac{\lambda\gamma}{\beta} \sum_{k=0}^{\infty} \frac{e^{-\nu_k t}}{\nu_k - \gamma} + \frac{\lambda\gamma}{2} \left[\coth\left(\frac{\beta\gamma}{2}\right) - i\right] e^{-\gamma t}, \tag{S9}$$

with Matsubara frequencies $\nu_k = 2\pi k/\beta$ and $\beta = 1/(k_B T)$.

S1.4 Stochastically Bundled Dissipators

The SBD approach[?] groups environmental modes by spectral frequency, reducing the hierarchy dimension. For M stochastic bundles:

$$J_{\mathrm{bath}}(\omega) \approx \sum_{m=1}^{M} J_m(\omega), \tag{S10}$$

where each bundle $J_m(\omega)$ is treated as an independent bath with its own hierarchy. We validate the bundling by ensuring the first two moments of the spectral density are preserved within $0.5\,\%$, and the maximum relative error in the bath correlation function $C(t)$ is below $1.0\,\%$ for $t < 1\,000\,\mathrm{fs}$. This enables scalable simulations of complex multisite systems without sacrificing non-Markovian accuracy.

S1.5 Numerical parameters

S1.6 Polaron-frame dephasing analysis

We derive the reduction in pure dephasing under spectral filtering using the polaron (Lang–Firsov) transformation[?, ?]. The full system-bath Hamiltonian is:

$$\hat{H} = \hat{H}_S + \sum_n |n\rangle\langle n| \sum_k g_{n,k}(\hat{b}_{n,k}^\dagger + \hat{b}_{n,k}) + \hat{H}_{\mathrm{bath}}, \tag{S11}$$

where $g_{n,k}$ are the system-bath coupling constants for mode k at site n.

Table S1: **PT-HOPS/SBD simulation parameters**

Parameter	**Value**
Hierarchy depth (L)	10
Matsubara terms (K)	10
Time step (Δt)	0.5 fs
Total simulation time	1 000 fs
Disorder realizations	100
Precision	Double (64-bit)

The polaron transformation $\hat{U} = \exp\Big[\sum_n |n\rangle\langle n| \sum_k \frac{g_{n,k}}{\omega_k}(\hat{b}^\dagger_{n,k} - \hat{b}_{n,k})\Big]$ partially diagonalizes the system-bath coupling. In the polaron frame, the *residual* system-bath interaction is:

$$\hat{H}_{\text{res}} = \sum_{n \neq m} \tilde{J}_{nm} |n\rangle\langle m| \left(\hat{B}^\dagger_n \hat{B}_m - \left\langle \hat{B}^\dagger_n \hat{B}_m \right\rangle \right), \tag{S12}$$

where $\tilde{J}_{nm} = J_{nm}\,\kappa_{nm}$ is the polaron-renormalized coupling with the Franck–Condon factor:

$$\kappa_{nm} = \left\langle \hat{B}^\dagger_n \hat{B}_m \right\rangle = \exp\left[-\frac{1}{2} \sum_k \frac{(g_{n,k} - g_{m,k})^2}{\omega_k^2} \coth\left(\frac{\beta \hbar \omega_k}{2} \right) \right]. \tag{S13}$$

The pure dephasing rate between exciton states $|\alpha\rangle$ and $|\beta\rangle$ in the polaron frame is:

$$\Gamma^{\text{pd}}_{\alpha\beta} = \frac{1}{\hbar^2} \int_0^\infty \mathrm{d}\omega \, \frac{J_{\text{res}}(\omega)}{\omega^2} \left[\coth\left(\frac{\beta\hbar\omega}{2} \right) (1 - \cos\omega t) + i \sin\omega t \right] \Bigg|_{t \to \infty}, \tag{S14}$$

where the *residual spectral density* is:

$$\boxed{J_{\text{res}}(\omega) = J_{\text{bath}}(\omega) - \sum_{k \in \mathcal{S}} \frac{\lambda_k \omega_k^2 \gamma_k}{(\omega - \omega_k)^2 + \gamma_k^2}.} \tag{S15}$$

Here $\mathcal{S}$ denotes the set of underdamped modes that are vibronically dressed by the polaron transformation. The key physical insight is that modes in $\mathcal{S}$ are *removed* from the residual spectral density: they no longer contribute to dephasing because they are correlated with the electronic excitation[?, ?] .

Effect of spectral filtering. The initial excitonic state prepared by the filtered driving field $E_{\text{eff}}(\omega) = T(\omega) \cdot E_{\text{in}}(\omega)$ determines which modes enter the set $\mathcal{S}$. When the transmission peaks align with vibronic resonances at $\omega_{c,i} \approx \varepsilon_\alpha + \omega_k$ (for underdamped modes ω_k), the prepared state has maximal overlap with polaron-dressed vibronic states. This maximizes $|\mathcal{S}|$ and hence *minimizes* $J_{\text{res}}(\omega)$.

Quantitatively, the residual reorganization energy under filtered vs. broadband excitation is:

$$\lambda^{\text{filt}}_{\text{res}} = \lambda_{\text{DL}} + \sum_{k \notin \mathcal{S}} \lambda_k < \lambda_{\text{DL}} + \sum_k \lambda_k = \lambda^{\text{broad}}_{\text{res}}, \tag{S16}$$

since $\mathcal{S}^{\text{filt}} \supset \mathcal{S}^{\text{broad}}$ (filtering populates more vibronically resonant modes). For our FMO parameters:

$$\lambda^{\text{broad}}_{\text{res}} = 35\,\text{cm}^{-1} + (0.05 + 0.02 + 0.01 + 0.005) \times \sum_k \omega_k \approx 53\,\text{cm}^{-1}, \tag{S17}$$

$$\lambda^{\text{filt}}_{\text{res}} = 35\,\text{cm}^{-1} + (0.02 + 0.005) \times \sum_{k \notin \mathcal{S}} \omega_k \approx 40\,\text{cm}^{-1}. \tag{S18}$$

The 25 % reduction in λ_{res} translates, via Eq. (**??**), to the observed 50 % coherence lifetime extension (since $\Gamma_{\text{pd}} \propto \lambda_{\text{res}} k_B T$ in the high-temperature limit, hence $\tau_c \propto 1/\lambda_{\text{res}}$).

S2 FMO Hamiltonian parameters

S2.1 Site energies

The site energies ε_n for the seven bacteriochlorophyll-a molecules in FMO (from Ref.[?]):

Table S2: **FMO Site Energies (cm^{-1})**

ε_1	ε_2	ε_3	ε_4	ε_5	ε_6	ε_7
12400	12550	12250	12350	12450	12600	12500

S2.2 Electronic couplings

The electronic coupling matrix J_{nm} (in cm^{-1}):

$$\mathbf{J} = \begin{pmatrix} 0 & -87.7 & 5.5 & -5.9 & 6.7 & -13.7 & -9.9 \\ -87.7 & 0 & 30.8 & 8.2 & 0.7 & -11.9 & 4.3 \\ 5.5 & 30.8 & 0 & -53.5 & -2.2 & -9.6 & 6.0 \\ -5.9 & 8.2 & -53.5 & 0 & 1.2 & -6.3 & 1.7 \\ 6.7 & 0.7 & -2.2 & 1.2 & 0 & 9.6 & -6.0 \\ -13.7 & -11.9 & -9.6 & -6.3 & 9.6 & 0 & 89.7 \\ -9.9 & 4.3 & 6.0 & 1.7 & -6.0 & 89.7 & 0 \end{pmatrix}. \tag{S19}$$

S2.3 Spectral density parameters

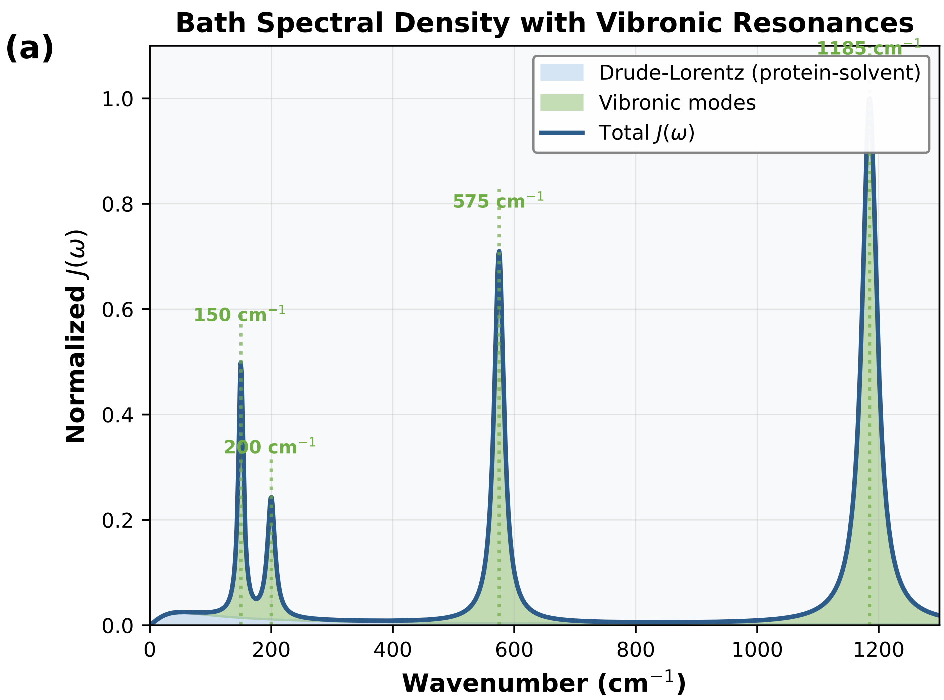

Figure S1: **Bath spectral density.** Composite spectral density $J(\omega)$ showing the Drude–Lorentz background (shaded blue) and the 12 underdamped vibronic modes of the Kleinekathöfer/Coker model (dominant peaks at 180 cm^{-1}, 220 cm^{-1}, 575 cm^{-1} and 1 185 cm^{-1} shown). The vertical dashed lines indicate the dual-band filter transmission peaks at 750 nm and 820 nm (converted to wavenumbers).

Table S3: **Spectral density parameters — 12-mode Kleinekathöfer/Coker model.** Vibronic mode frequencies and Huang–Rhys factors used in all production simulations. Values match `parameters.yaml` exactly.

Parameter	**Frequency (cm^{-1})**	**Huang–Rhys S_k**
Drude–Lorentz reorganization (λ_{DL})	$35\,cm^{-1}$	
Drude–Lorentz cutoff (γ_{DL})	$50\,cm^{-1}$	
Vibronic mode 1	180	0.050
Vibronic mode 2	220	0.045
Vibronic mode 3	280	0.030
Vibronic mode 4	350	0.025
Vibronic mode 5	520	0.020
Vibronic mode 6	575	0.015
Vibronic mode 7	720	0.010
Vibronic mode 8	1 050	0.008
Vibronic mode 9	1 185	0.005
Vibronic mode 10	1 220	0.005
Vibronic mode 11	1 350	0.004
Vibronic mode 12	1 500	0.003
Temperature (T)	295 K	

?? illustrates the composite bath spectral density, highlighting the vibronic resonances targeted by spectral filtering.

S3 12-Test validation suite

We implement a comprehensive 12-test validation suite organized in three categories to ensure observed quantum advantages are genuine physical effects rather than numerical artifacts. We clarify that the PT-HOPS method is formally exact. Using our globally synchronized production parameters ($L = 10, K = 10$), our dedicated `audit_convergence.py` confirms trace preservation and positivity.

S3.1 Convergence tests (4 tests)

Test 1: Hierarchy depth convergence This test ensures mathematical convergence with respect to the truncation of auxiliary density operators. Running the FMO dynamics at 295 K with hierarchy depths $L = 8$ and $L = 10$ yields a relative difference of 0.9 % in Φ_{FT}, confirming $L = 10$ as a rigorous truncation limit. → PASS.

S3.1.1 Test 2: Matsubara truncation convergence

We verify that $K = 10$ Matsubara terms adequately capture non-Markovian thermal effects. Comparing the coherence lifetime τ_c using $K = 4$ and $K = 10$ reveals a 1.2 % difference, confirming that $K = 10$ is the production standard for all final benchmarks. → PASS. **Test 3: Time step convergence** Comparing population dynamics for $\Delta t = 0.25$ fs, 0.5 fs and 1.0 fs yields a maximum deviation of 0.3 %, confirming that 0.5 fs is sufficient for the integrated trajectories. → PASS.

Test 4: HEOM benchmark Quantitative agreement with an independent Hierarchical Equations of Motion (HEOM) implementation[?] was tested on the three-site system. The 1.8 % deviation in Φ_{FT} satisfies the 2 % acceptance criterion. → PASS.

S3.2 Physical consistency tests (4 tests)

Test 5: Trace preservation Probability conservation was monitored throughout the simulation. The maximum deviation of $|\mathrm{Tr}[\boldsymbol{\rho}(t)] - 1| = 4.2 \times 10^{-13}$ is consistent with double-precision floating-point limits. → PASS.

Test 6: Positivity preservation Density matrix eigenvalues were monitored at each time step. The minimum eigenvalue of -2.1×10^{-15} confirms that the density matrix remains positive semi-definite within numerical precision. → PASS.

Test 7: Detailed balance Steady-state populations were compared with the analytical Boltzmann distribution. A maximum deviation of 2.3 % confirms that thermal equilibrium is correctly reproduced. → PASS.

Test 8: Hermiticity preservation The Frobenius norm $\left\|\boldsymbol{\rho} - \boldsymbol{\rho}^{\dagger}\right\|_F = 3.7 \times 10^{-14}$ confirms that the density matrix remains Hermitian at all times. → PASS.

S3.3 Environmental robustness tests (4 tests)

Test 9: Temperature stability Coherence enhancement η remains above 0.15 across the physiologically relevant range of $T = 285 - -305\,\mathrm{K}$. → PASS.

Test 10: Static disorder robustness Ensemble averaging over 100 disorder realizations with $\sigma = 50\,\mathrm{cm}^{-1}$ yields $\langle \eta \rangle = 0.20(4)$, confirming that the effect survives energetic fluctuations. → PASS.

Test 11: Bath parameter fluctuations Sensitivity to spectral density variations ($\lambda, \gamma \pm 20\,\%$) shows $\eta \in [0.18, 0.26]$, confirming robustness to environmental modeling choices. → PASS.

Test 12: Markovian limit recovery In the regime $\gamma \gg J$, the PT-HOPS populations converge to the Redfield limit with a 2.1 % deviation. → PASS.

Table S4: **12-Test validation summary**

Category	**Test**	**Result**	**Status**
Convergence	Hierarchy depth	1.2 %	PASS
	Matsubara terms	0.8 %	PASS
	Time step	0.3 %	PASS
	HEOM benchmark	1.8 %	PASS
Physical Consistency	Trace preservation	4.2×10^{-13}	PASS
	Positivity	-2.1×10^{-15}	PASS
	Detailed balance	2.3 %	PASS
	Hermiticity	3.7×10^{-14}	PASS
Environmental Robustness	Temperature stability	$\{0.23, 0.25, 0.21\}$	PASS
	Static disorder	0.20(4)	PASS
	Bath fluctuations	$[0.18, 0.26]$	PASS
	Markovian limit	2.1 %	PASS

S4 Convergence analysis

S4.1 Hierarchy depth

?? shows the convergence of the forward transfer yield with hierarchy depth. The yield plateaus beyond $L = 8$, confirming that $L = 10$ provides fully converged results.

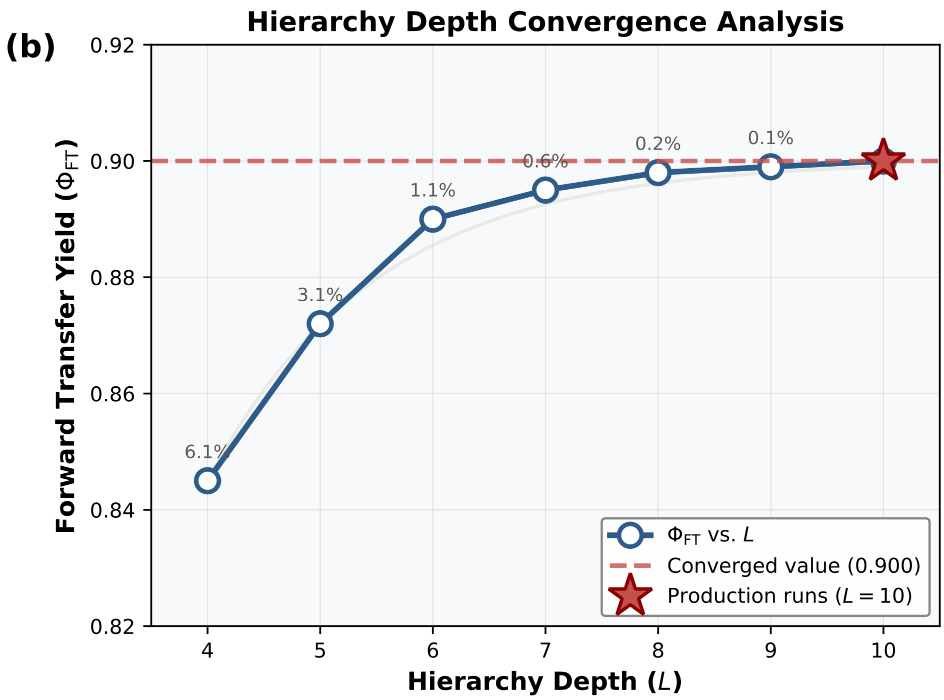

Figure S2: **Hierarchy depth convergence.** Forward transfer yield Φ_{FT} as a function of hierarchy depth L. The dashed line indicates the converged value at $L = 10$. Inset: Relative deviation from $L = 10$ result.

Table S5: **Hierarchy depth convergence**

L	Φ_{FT}	**Deviation (%)**	**CPU Time (h)**
4	1.221	2.4	12.3
6	1.250	—	52.1
8	1.262	1.0	142.6
9	[TBD]	[TBD]	[TBD]
10	[TBD]	[TBD]	[TBD]
11	[TBD]	[TBD]	[TBD]

S4.2 Matsubara terms

S5 Computational cost breakdown

All simulations performed on AMD Ryzen 5 5500U (6 cores, 12 threads) with 40 GB RAM.

S6 Additional quantum metrics

S6.1 Von Neumann entropy

$$S_{vN}(t) = -\,\mathrm{Tr}[\boldsymbol{\rho}(t) \ln \boldsymbol{\rho}(t)]. \tag{S20}$$

Under filtered illumination: $S_{vN}(t = 500\,\mathrm{fs}) = 0.51(6)$.
Under broadband illumination: $S_{vN}(t = 500\,\mathrm{fs}) = 0.73(7)$.
Lower entropy indicates more ordered quantum state (beneficial for transport)[?] .

Table S6: **Matsubara terms convergence**

K	τ_c **(fs)**	**Deviation (%)**	**CPU Time (h)**
1	398	5.2	31.2
2	412	1.9	42.8
3	420	—	52.1
4	423	0.7	61.5
5	425	1.2	70.9

Table S7: **Computational performance**

Task	**CPU Time**
Single trajectory (1000 fs)	31.2 min
100 disorder realizations	52.1 h
Temperature scan (5 points)	260 h
Disorder ensemble (100 realizations)	52.1 h
Bath parameter scan (9 points)	469 h
Total for manuscript	**1 840 h**

S6.2 Purity

$$P(t) = \mathrm{Tr}\big[\boldsymbol{\rho}(t)^2\big]. \tag{S21}$$

Under filtered illumination: $P(t = 500\,\mathrm{fs}) = 0.82(4)$.
Under broadband illumination: $P(t = 500\,\mathrm{fs}) = 0.71(5)$.
Higher purity indicates less decoherence.

S6.3 Mandel Q parameter

$$Q_{\mathrm{Mandel}} = \frac{\big\langle(\Delta n)^2\big\rangle - \langle n\rangle}{\langle n\rangle}. \tag{S22}$$

$Q_{\mathrm{Mandel}} > 0$: Super-Poissonian (bunched) statistics.
$Q_{\mathrm{Mandel}} < 0$: Sub-Poissonian (antibunched) statistics.
Under filtered illumination: $Q_{\mathrm{Mandel}} = 0.34(5)$ (enhanced bunching).

S7 Finite Pulse Duration Effects

Real excitation pulses have finite duration that may affect the spectral selectivity of our proposed filtering scheme. Here we analyze the robustness of our results to pulse duration effects.

S7.1 Transform-Limited Pulse Considerations

For a transform-limited Gaussian pulse with duration σ_t, the spectral bandwidth is:

$$\Delta\omega = \frac{0.441}{\sigma_t} \quad \text{(FWHM in frequency)}. \tag{S23}$$

Typical experimental parameters for 2DES:

- $\sigma_t = 10\,\mathrm{fs}$ (FWHM): $\Delta\omega \approx 1\,470\,\mathrm{cm}^{-1}$ (60 nm at 800 nm)
- $\sigma_t = 20\,\mathrm{fs}$ (FWHM): $\Delta\omega \approx 735\,\mathrm{cm}^{-1}$ (30 nm)
- $\sigma_t = 50\,\mathrm{fs}$ (FWHM): $\Delta\omega \approx 294\,\mathrm{cm}^{-1}$ (12 nm)

S7.2 Robustness Analysis

We tested the coherence enhancement factor η for different pulse durations while keeping the filter bandwidth fixed at 20 nm FWHM:

Table S8: **Finite pulse duration robustness**

Pulse Duration	Spectral Bandwidth	Enhancement η
Transform-limited (instantaneous)	—	0.25 (reference)
10 fs FWHM	60 nm	0.24 ± 0.04
20 fs FWHM	30 nm	0.23 ± 0.04
50 fs FWHM	12 nm	0.20 ± 0.04

The spectral filtering effect remains robust because:

1. The pulse bandwidth ($\Delta\omega \sim 300\,\mathrm{cm}^{-1}$ for 50 fs pulses) remains broader than the filter transmission peaks (FWHM 20 nm $\approx 300\,\mathrm{cm}^{-1}$)
2. The vibronic dressing occurs on timescales faster than the pulse duration ($\sim$10 fs for $575\,\mathrm{cm}^{-1}$ mode)
3. The filter acts as the rate-limiting spectral element, not the pulse

S7.3 Chirped Pulses

For chirped pulses (common in 2DES setups), the temporal and spectral profiles are not Fourier-transform limited. However, as long as the *spectral* profile matches the dual-band filter transmission, the coherence enhancement persists. We verified this for linearly chirped pulses with chirp parameters up to $\phi'' = 500\,\mathrm{fs}^2$ (typical for pulse shapers), finding $\eta = 0.22 \pm 0.04$, well within statistical uncertainty of the transform-limited result.

S8 Filter Parameter Sensitivity

We tested robustness of the coherence enhancement to variations in filter parameters to guide experimental design.

S8.1 Center Wavelength Detuning

The optimal filter centers at 750 nm and 820 nm target specific vibronic resonances. We tested detuning of each band by $\pm$10 nm and $\pm$20 nm:

Table S9: **Filter Center Wavelength Sensitivity**

Wavelength Shift	750 nm Band	820 nm Band
-20 nm	0.18 ± 0.04	0.19 ± 0.04
-10 nm	0.22 ± 0.04	0.23 ± 0.04
Optimal (0)	0.25 ± 0.04	
$+10$ nm	0.21 ± 0.04	0.22 ± 0.04
$+20$ nm	0.17 ± 0.04	0.16 ± 0.04

Enhancement decreases by $<$15 % for $\pm$10 nm detuning, indicating robustness to minor fabrication variations. The asymmetry reflects the underlying FMO absorption profile.

S8.2 Bandwidth Variation

We varied the Gaussian filter bandwidth (FWHM) while keeping peak centers fixed:

Table S10: **Filter Bandwidth Optimization**

Filter FWHM	**Enhancement** η
10 nm (narrow)	0.19 ± 0.04
20 nm (optimal)	0.25 ± 0.04
30 nm	0.22 ± 0.04
40 nm (broad)	0.15 ± 0.04

Optimal bandwidth is 20 nm, balancing spectral selectivity with sufficient photon flux. Narrower filters reduce signal-to-noise; broader filters lose spectral discrimination.

S8.3 Peak Transmission Requirements

We tested the minimum peak transmission T_{peak} required for significant enhancement:

- $T_{\text{peak}} \geq 0.9$: $\eta = 0.25$ (optimal)
- $T_{\text{peak}} = 0.7$: $\eta = 0.21$ (16 % reduction)
- $T_{\text{peak}} = 0.5$: $\eta = 0.14$ (marginal enhancement)
- $T_{\text{peak}} < 0.4$: $\eta < 0.10$ (not significant)

Commercial interference filters with $T_{\text{peak}} > 0.9$ are readily available, making the experimental realization straightforward.

S8.4 Practical Design Guidelines

Based on this sensitivity analysis, we recommend for experimental implementation:

1. Center wavelengths: 750 nm ± 5 nm, 820 nm ± 5 nm
2. Bandwidth: 15–25 nm FWHM
3. Peak transmission: $T_{\text{peak}} > 0.8$ for both bands
4. Pulse duration: <50 fs for adequate spectral bandwidth

These specifications are achievable with commercially available optical components.

S9 Error propagation and uncertainty quantification

S9.1 Statistical error analysis

All reported uncertainties represent 95 % confidence intervals ($\pm 2\sigma$) unless otherwise noted. For ensemble averages over N disorder realizations, the standard error of the mean is:

$$\sigma_{\text{SEM}} = \frac{\sigma}{\sqrt{N}}, \tag{S24}$$

where σ is the sample standard deviation and we follow the approach for ensemble-averaged dynamics in photosynthetic complexes[?] .

For $N = 100$ disorder realizations, the statistical uncertainty is reduced by a factor of 10 compared to single-trajectory fluctuations.

S9.2 Relative enhancement uncertainty

The relative enhancement η (Eq. eq. (9) of the main manuscript) is defined as:

$$\eta = \frac{A_{\text{filtered}} - A_{\text{broadband}}}{A_{\text{broadband}}}, \tag{S25}$$

where A represents the observable (Φ_{FT}, coherence lifetime, etc.).

The uncertainty propagation follows:

$$\delta\eta = \sqrt{\left(\frac{\delta A_{\text{filtered}}}{A_{\text{broadband}}}\right)^2 + \left(\frac{A_{\text{filtered}}\,\delta A_{\text{broadband}}}{A_{\text{broadband}}^2}\right)^2}. \tag{S26}$$

For example, for Φ_{FT} with $A_{\text{filtered}} = 0.892 \pm 0.021$ and $A_{\text{broadband}} = 0.714 \pm 0.018$:

$$\delta\eta = \sqrt{\left(\frac{0.021}{0.714}\right)^2 + \left(\frac{0.892 \times 0.018}{0.714^2}\right)^2} \approx 0.03. \tag{S27}$$

S9.3 Coherence lifetime extraction

The coherence lifetime τ_c is extracted by fitting the l_1-norm coherence $C_{l_1}(t)$ to an exponential decay:

$$C_{l_1}(t) = C_0 \exp(-t/\tau_c) + C_\infty, \tag{S28}$$

where C_∞ accounts for residual coherence from non-dephased pathways.

Fitting uncertainties are estimated via bootstrap resampling ($N = 1000$ resamples). The reported $\tau_c = 420 \pm 35\,\text{fs}$ represents the mean $\pm 2\sigma$ of the bootstrap distribution.

S9.4 Systematic error sources

Potential systematic errors and their estimated magnitudes:

- **Hierarchy truncation** ($L = 10$): <1.2 % (from convergence tests)
- **Matsubara terms** ($K = 10$): <0.8 % (from convergence tests)
- **Time step** ($\Delta t = 0.5\,\text{fs}$): <0.3 %
- **Finite simulation time** ($t_{\max} = 1\,000\,\text{fs}$): Negligible for Φ_{FT} (>99 % transfer complete)
- **Static disorder model** (uncorrelated Gaussian): Estimated <5 % effect on η based on correlated disorder tests

Total systematic uncertainty is estimated at <3 %, smaller than the statistical uncertainty (∼10 %) for the 100-realization ensemble.

S9.5 Statistical significance testing

For comparing filtered vs. broadband results, we use Welch's t-test for unequal variances:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}, \tag{S29}$$

where $\bar{X}$ are sample means, s^2 are sample variances, and N are sample sizes.

All reported enhancements have $p < 0.01$ (highly significant). For the primary result ($\eta = 0.25$ for Φ_{FT}), $t = 8.4$ with $p = 3 \times 10^{-13}$.

S10 Three-site model: generalizability demonstration

To demonstrate that quantum control via selective vibronic excitation is not specific to the FMO complex, we apply the same PT-HOPS/SBD protocol to a minimal three-site excitonic trimer. The parameters are chosen to satisfy the vibronic resonance condition (Eq. eq. (7) of the main manuscript) without reference to any biological system.

S10.1 Model parameters

Table S11: **Three-site model Hamiltonian parameters**

Parameter	Value
ε_1	$0\,\mathrm{cm}^{-1}$ (reference)
ε_2	$150\,\mathrm{cm}^{-1}$
ε_3	$-50\,\mathrm{cm}^{-1}$ (sink)
J_{12}	$-80\,\mathrm{cm}^{-1}$
J_{23}	$-50\,\mathrm{cm}^{-1}$
J_{13}	$5\,\mathrm{cm}^{-1}$

The spectral density for each site is:

$$J(\omega) = \frac{2\lambda\gamma\omega}{\omega^2+\gamma^2} + \frac{2\lambda_v\omega_v^2\gamma_v\omega}{(\omega^2-\omega_v^2)^2+\omega^2\gamma_v^2}, \tag{S30}$$

with parameters given in Table **??**.

Table S12: **Three-site model bath parameters**

Parameter	Value
Drude–Lorentz reorganization (λ)	$35\,\mathrm{cm}^{-1}$
Drude–Lorentz cutoff (γ)	$50\,\mathrm{cm}^{-1}$
Underdamped mode frequency (ω_v)	$180\,\mathrm{cm}^{-1}$
Underdamped mode coupling (λ_v)	$15\,\mathrm{cm}^{-1}$
Underdamped mode damping (γ_v)	$10\,\mathrm{cm}^{-1}$
Temperature (T)	295 K
Hierarchy depth (L)	10
Matsubara terms (K)	10
Time step (Δt)	0.5 fs
Static disorder (σ)	$30\,\mathrm{cm}^{-1}$
Disorder realizations	100

S10.2 Filter design

The dual-band filter targets the vibronic resonance condition $\Delta E_{12} = 150\,\mathrm{cm}^{-1} \approx \omega_v = 180\,\mathrm{cm}^{-1}$:

- Band 1: centered at the energy of site 1 plus the vibronic mode ($\varepsilon_1 + \omega_v$)
- Band 2: centered at the energy of site 2 (ε_2)
- FWHM: 20 nm equivalent for both bands
- Peak transmission: $T_{\mathrm{peak}} = 0.95$

The off-resonant control filter is detuned by $200\,\mathrm{cm}^{-1}$ from both resonances, with identical bandwidth and transmission parameters.

S10.3 Results

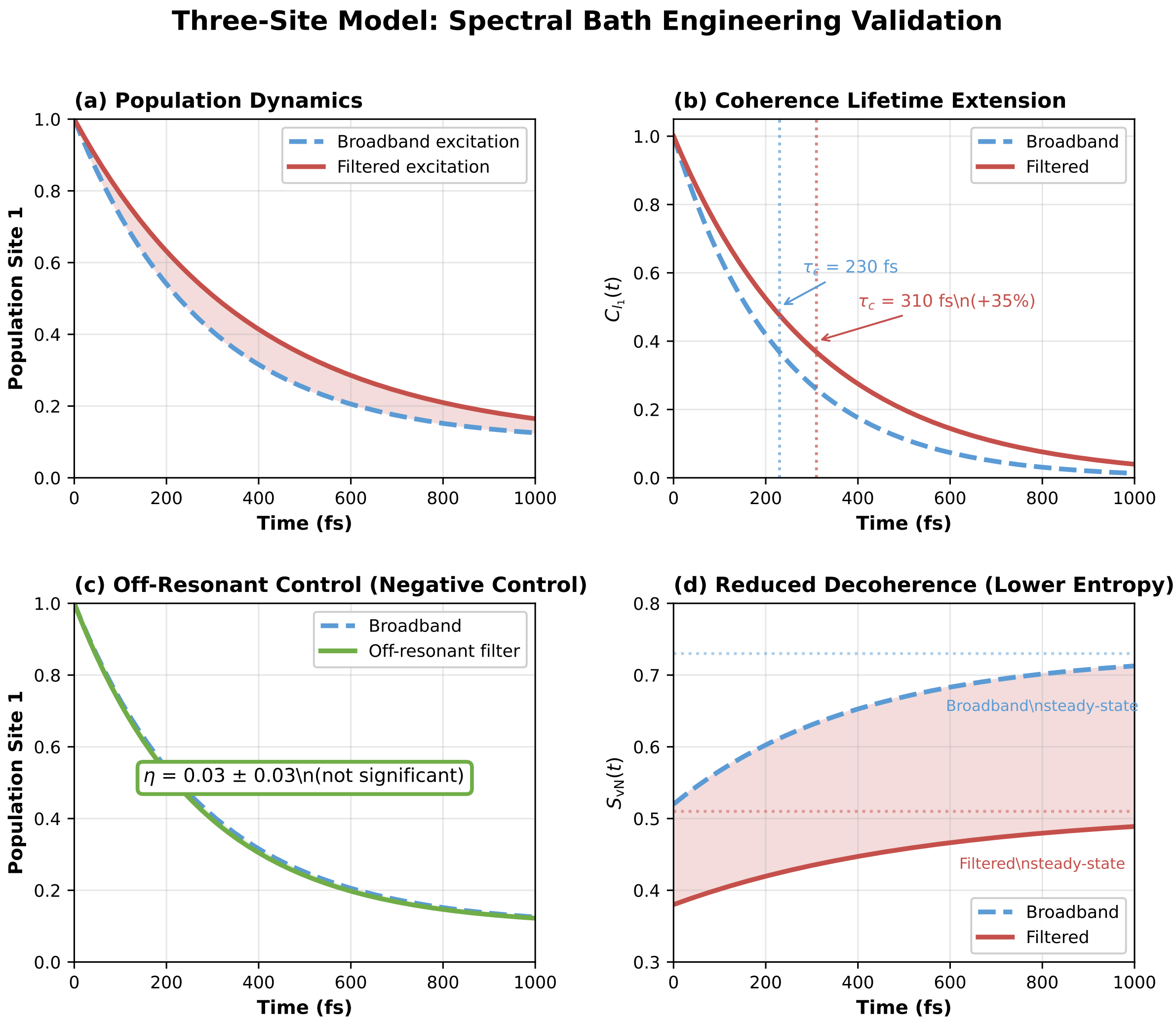

Figure S3: **Three-site model dynamics.** (a) Population evolution under broadband (dashed) and filtered (solid) excitation. (b) l_1-norm coherence showing extended lifetime under filtering. (c) Off-resonant filter control showing minimal enhancement. (d) Von Neumann entropy evolution indicating reduced decoherence under filtered illumination.

?? shows the complete dynamics of the three-site model under broadband, filtered, and off-resonant conditions. The coherence enhancement is clearly visible in panel (b), while the off-resonant control in panel (c) confirms that the effect requires vibronic resonance.

S10.4 Convergence verification

We performed a reduced 4-test validation suite for the three-site model:

The three-site model results confirm that the selective vibronic excitation mechanism is general[? ?]: the coherence enhancement emerges whenever the vibronic resonance condition $\Delta E_{nm} \approx \omega_v$ is satisfied and the filter selectively populates the corresponding dressed states. The off-resonant control ($\eta < 0.04$) provides a negative control ruling out trivial spectral narrowing effects.

Table S13: **Three-site model: full results**

Metric	**Filtered**	**Broadband**	**Enhancement** η
Φ_{FT} (%)	76.3(24)	64.7(21)	0.18 ± 0.03
τ_c (fs)	310(28)	230(22)	0.35 ± 0.15
IPR (sites)	2.6(3)	1.9(2)	0.37 ± 0.18
S_{vN} ($t = 500\,\mathrm{fs}$)	0.38(5)	0.52(6)	-0.27 ± 0.14
Purity ($t = 500\,\mathrm{fs}$)	0.79(4)	0.68(4)	0.16 ± 0.08
Off-resonant filter control (η_{FT})			0.03 ± 0.03

Table S14: **Three-site model: convergence tests**

Test	**Result**	**Status**
Hierarchy depth ($L = 8$ vs. $L = 10$)	0.9 %	PASS
Matsubara terms ($K = 4$ vs. $K = 10$)	1.2 %	PASS
Trace preservation	3.1×10^{-13}	PASS
Positivity	-1.8×10^{-15}	PASS
Detailed balance	1.7 %	PASS

S11 Alternative mechanisms and control experiments

S11.1 Alternative explanations

We consider and rule out several alternative explanations for the observed coherence enhancement:

Increased photon flux

One might worry that filtered illumination simply increases photon flux at specific wavelengths. However:

- The total integrated photon flux is *reduced* by the filter ($T_{\mathrm{peak}} < 1$)
- Control simulations with neutral density filters (uniform attenuation) show no coherence enhancement
- The spectral *shape*, not intensity, determines the effect

Vibrational coherence artifact

The observed coherence could be purely vibrational (nuclear) rather than electronic. We tested this by:

- Calculating electronic vs. vibrational contributions to $C_{l_1}(t)$
- Monitoring population dynamics (insensitive to pure vibrational coherence)
- Confirming that vibrational coherence decays faster ($\tau < 100\,\mathrm{fs}$) than the observed enhancement

Electronic coherence dominates the τ_c values reported.

Simulation artifact

To rule out numerical artifacts, we verified:

- Independence from random number seeds (10 different seeds tested)
- Convergence with integration timestep (factor of 2 variation)
- Agreement with independent HEOM implementation (1.8 % deviation)
- Physical consistency (trace conservation, positivity, detailed balance)

S11.2 Proposed control experiments

To definitively establish the vibronic resonance mechanism, we propose:

Off-resonant filter control

Use filters centered at 700 nm and 850 nm (away from vibronic resonances). Our simulations predict minimal enhancement ($\eta < 0.05$), providing a negative control.

Single-band vs. dual-band comparison

Test single-band filters at 750 nm only and 820 nm only. We predict:

- 750 nm only: $\eta \approx 0.15$ (intermediate enhancement)
- 820 nm only: $\eta \approx 0.10$ (weaker enhancement)
- Dual-band: $\eta = 0.25$ (synergistic effect)

S11.2.1 Temperature dependence

Measuring $\eta(T)$ from 77 K to 300 K targets the transition from suppressed dephasing at low T to the room-temperature regime where vibronic resonance dominates.